

Spec — Operations Monitoring: Mac + EC2 Division of Labor

Created: 2026-06-22 — Claude.ai + Grok Status: spec_ready — build after Block B German batch Priority: Medium

Context

EC2 is production and has no operational monitoring. Mac has informal monitoring via OpenClaw. This spec adds production-grade monitoring on EC2 and clarifies responsibilities between OpenClaw, CoS, and the system bot.

Core principle for this phase: CoS detects problems and escalates to Robert. Robert decides what action to take. CoS does not self-heal or restart services. Human in control of all production changes.

Division of Labor

Agent	Scope	Nature
OpenClaw	Independent system-wide checks — spans beyond minimoi	Proactive, autonomous, no hard dependency
CoS	minimoi-specific EC2 operational health	Detection + escalation only
minimoi_system_bot	Alert delivery channel	Sends notifications, not the checker

Overlap between OpenClaw and CoS is intentional — resilience over elegance.

Part 1 — EC2 Health Monitoring (CoS task)

Periodic task in `chief_of_staff.py`, runs on EC2 only via `is_production()` guard, alerts via `minimoi_system_bot`.

Two-layer tooling:

Layer 1 — AWS-native (CloudWatch):

- EBS disk utilization

- Instance CPU and memory
- Network in/out Already collected by AWS at no extra cost. Query via `boto3`.

Layer 2 — Portable Linux tools (work on any future machine):

- `docker ps` — container status
- `df -h` — disk usage
- `free -m` — memory pressure
- `curl localhost:{port}/health` — Flask app responding

Both layers used. AWS metrics for infrastructure visibility, Linux tools for application health. Neither alone is sufficient.

Checks every 30 minutes:

- All four containers running: portal, curator, german, postgres
- Disk < 80% (CloudWatch EBS metric + `df -h` cross-check)
- Memory < 85%
- All three `/health` endpoints return 200

Alert conditions (CoS escalates, does not fix):

- Container not running → immediate alert
- Disk > 80% → alert with usage and largest directories
- Memory > 85% → alert with top consumers
- Any `/health` non-200 → alert with app name and status

Alert format — informative and actionable:

```
⚠ EC2 Health Alert
Container: minimoi-curator is not running
Last seen: 14 minutes ago
To diagnose: docker logs minimoi-curator
To restart: docker-compose up -d curator
```

Robert sees the problem, context, and the exact command. CoS does not run the command.

Implementation:

```
python
```

```
def check_ec2_health():
    """
    Runs on EC2 only. Detects and escalates. Does not fix.
    """
    if not is_production():
        return

    # Layer 2: subprocess for docker ps, df, free, curl /health
    # Layer 1: boto3 for CloudWatch metrics
    # Alert via get_system_token() if thresholds breached
```

No new SSM parameters needed. Uses existing EC2 IAM role.

Part 2 — Mac Monitoring Cleanup

- Remove stale `rvsopenbot` references from Mac monitoring code
 - Confirm all Mac alerts route through `get_system_token()`
 - Confirm `minimoi_cmd_bot` and `com.user.telegram-feedback-bot` launchd jobs are unloaded and disabled on Mac
 - Update any hardcoded keyring references that survived A3
-

Part 3 — OPERATIONS.md Update

Update `docs/OPERATIONS.md` with:

- Two-node architecture (EC2 = production, Mac = standby)
 - What CoS monitors on EC2 and at what interval
 - What OpenClaw monitors independently
 - How to manually check EC2 health (commands included)
 - How to restart containers on EC2 if CoS alerts
 - Alert routing (all alerts → `minimoi_system_bot`)
 - Colima maintenance procedures for Mac
-

Definition of Done

- ☐ CoS health check task running on EC2 every 30 minutes
- ☐ Alerts fire via `minimoi_system_bot` for all four conditions

- ☐ Verified: stop a container on EC2, alert arrives within 30 min
- ☐ Stale rvsopenbot references removed from Mac monitoring
- ☐ Old Mac launchd jobs unloaded and disabled
- ☐ docs/OPERATIONS.md updated for two-node architecture

Commit sequence

1. Add `check_ec2_health()` to `chief_of_staff.py` (`is_production()` guard)
2. Mac monitoring cleanup – rvsopenbot references, retired launchd jobs
3. Update `docs/OPERATIONS.md`

Commit message

Operations: EC2 health monitoring via CoS, Mac cleanup, OPERATIONS.md updated for two-node architecture.

*Spec · 2026-06-22 · Claude.ai Register in design_log as spec_ready Roadmap companion:
docs/OPERATIONS_ROADMAP.md*